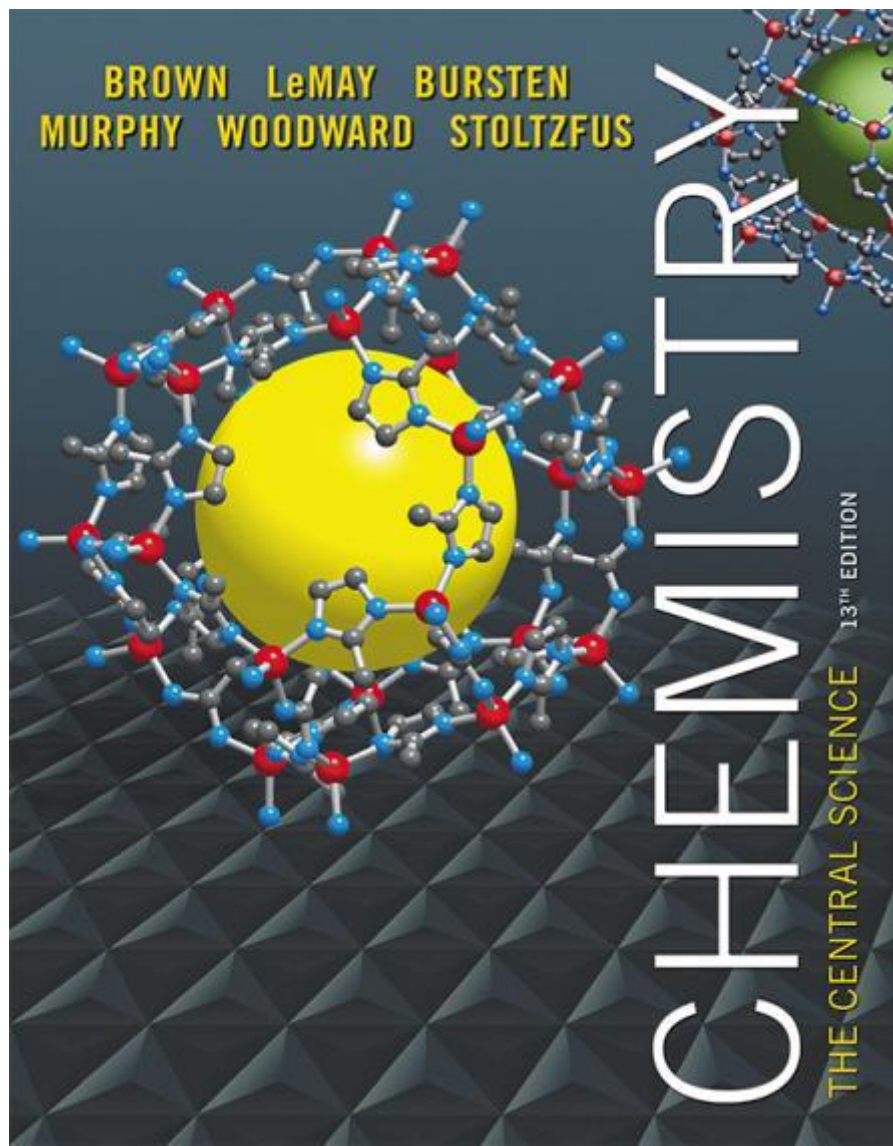


Chapter 3. Intermolecular Forces

分子间作用力



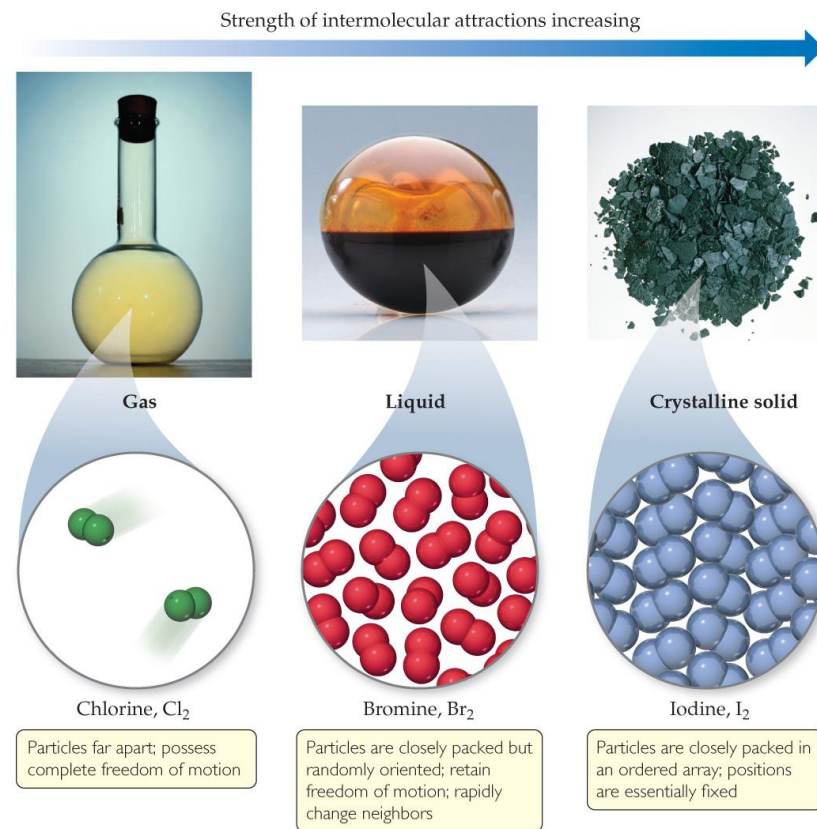
Reference:

Chapter 11

Liquids and Intermolecular Forces

States of Matter

- The fundamental difference between states of matter is the strength of the **intermolecular forces of attraction**.
- Stronger forces bring molecules closer together.
- Solids and liquids are referred to as the ***condensed phases***.



浓缩相（指物质存在的凝聚相态，如液态和固态）

Differences in the States of Matter

Table 11.1 Some Characteristic Properties of the States of Matter

Gas	Assumes both volume and shape of its container Expands to fill its container Is compressible Flows readily Diffusion within a gas occurs rapidly
Liquid	Assumes shape of portion of container it occupies Does not expand to fill its container Is virtually incompressible Flows readily Diffusion within a liquid occurs slowly
Solid	Retains own shape and volume Does not expand to fill its container Is virtually incompressible Does not flow Diffusion within a solid occurs extremely slowly

*The atoms in a solid are able to vibrate in place. As the temperature of the solid increases, the vibrational motion increases.

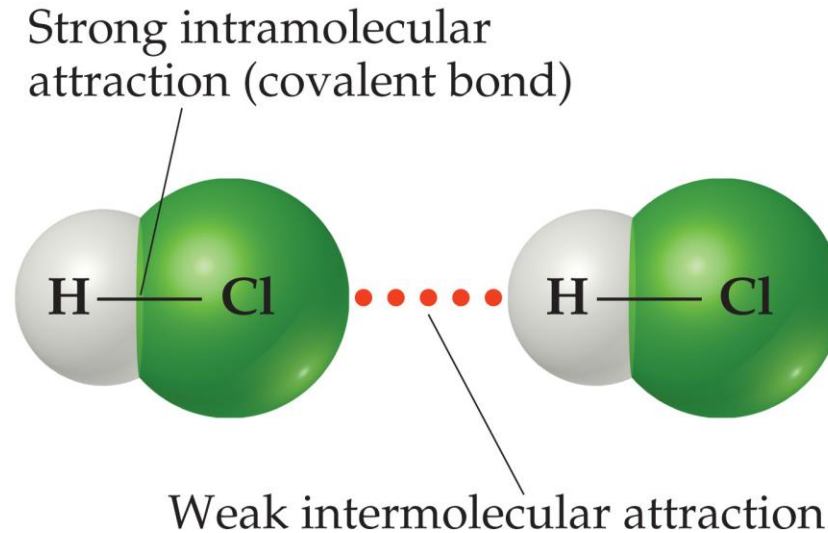
Which State of Matter?

- The answer to this question largely relies on the
 - balance between the 动能 kinetic energies of the particles.
 - interparticle energies of attraction.

Table 11.2 Comparing Kinetic Energies and Energies of Attractions for States of Matter

Gas	Kinetic energies $\gg$ energies of attraction
Liquid	Comparable kinetic energies and energies of attraction
Solid	Energies of attraction $\gg$ kinetic energies

Intermolecular Forces



- The attractions between molecules are not nearly as strong as the intramolecular attractions (bonds) that hold compounds together.
 - Many physical properties reflect **intermolecular forces**, like boiling points, melting points, viscosity, surface tension, and capillary action.
- 表面张力 毛细作用

Types of Intermolecular Force

- Weakest to strongest forces:
 - dispersion forces (or London dispersion forces, or *van der Waals forces*)
 - dipole–dipole forces
 - hydrogen bonding (a special dipole–dipole force)
 - ion–dipole forces

Dispersion Forces

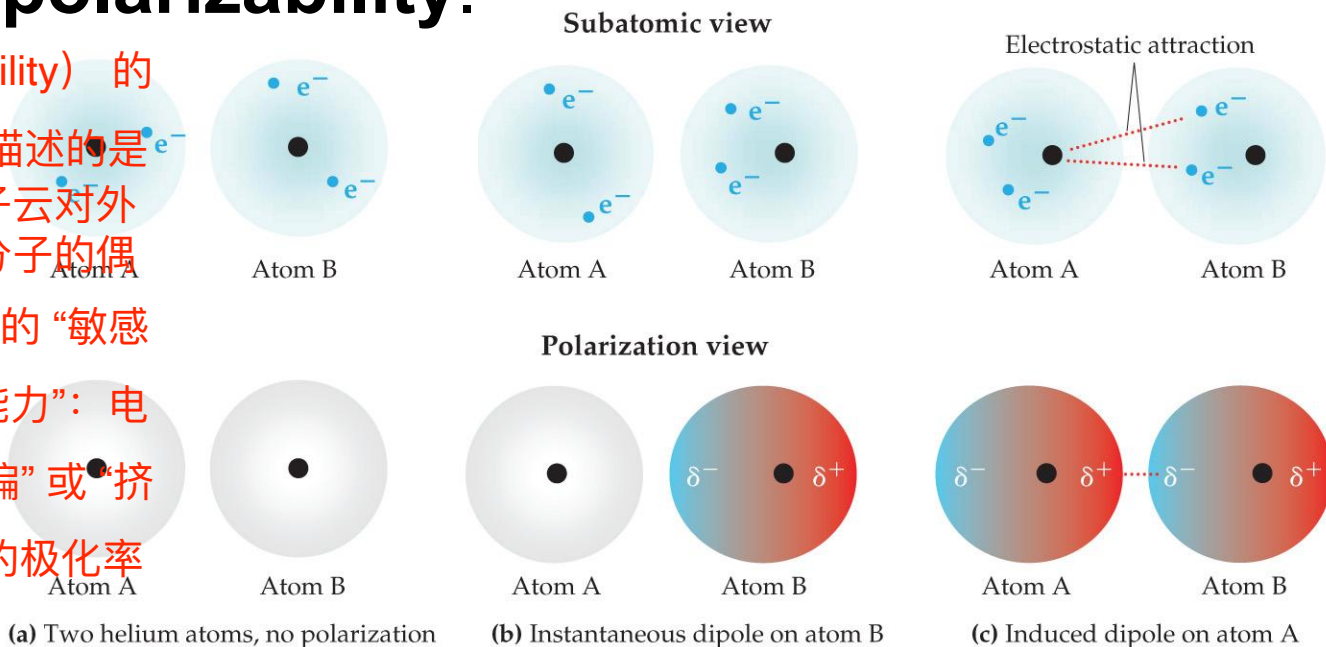
非极性氢原子

暂时极化

- A nonpolar helium atom can be temporarily (a fraction of femtosecond) polarized to allow dispersion force to form.
- The tendency of an electron cloud to distort is called its **polarizability**.

变形

极化率 (polarizability) 的核心本质——它描述的是分子或原子的电子云对外界电场（如其他分子的偶极、离子的电荷）的“敏感程度”或“易形变能力”：电子云越容易被“拉偏”或“挤压变形”，该粒子的极化率就越大。

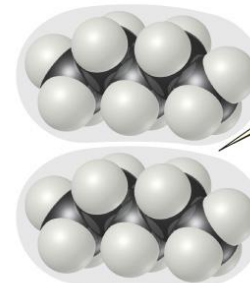


<https://www.khanacademy.org/science/ap-chemistry-beta/x2eef969c74e0d802:intermolecular-forces-and-properties/x2eef969c74e0d802:intermolecular-forces/v/london-dispersion-forces>

Factors Which Affect Amount of Dispersion Force in a Molecule

- **number of electrons** in an atom (more electrons, more dispersion force)
- **size** of atom or molecule/molecular weight
- **shape** of molecules with similar masses (more compact, less dispersion force)

Linear molecule—larger surface area enhances intermolecular contact and increases dispersion force



n-Pentane (C_5H_{12})
bp = 309.4 K

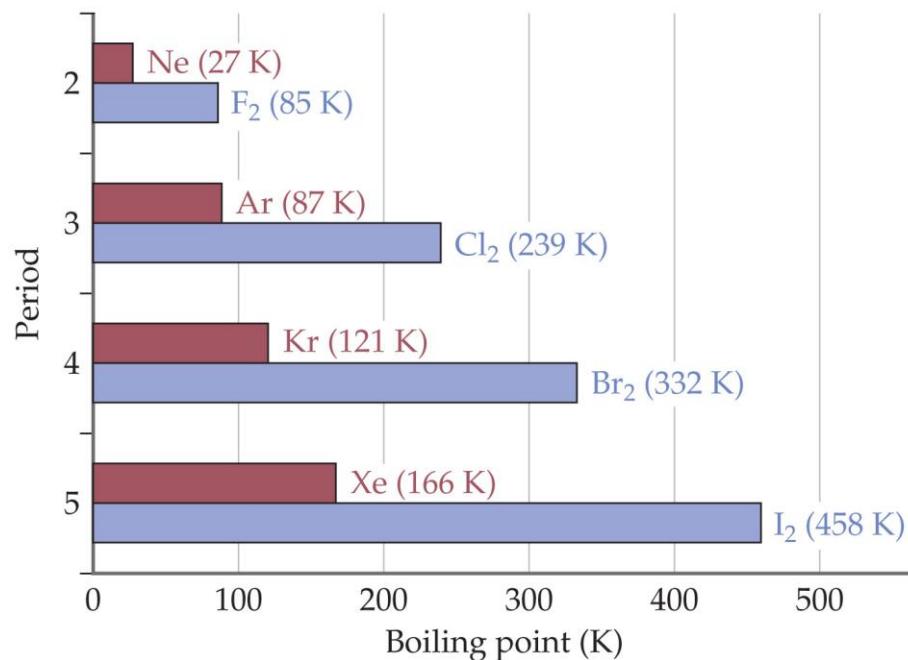
Spherical molecule—smaller surface area diminishes intermolecular contact and decreases dispersion force



Neopentane (C_5H_{12})
bp = 282.7 K

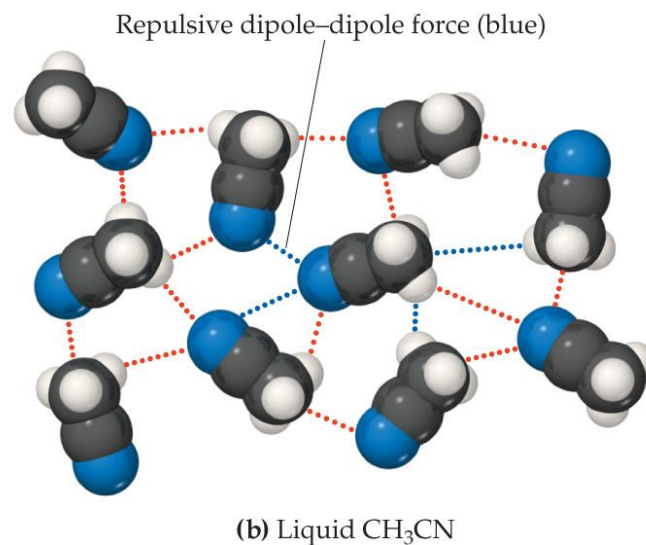
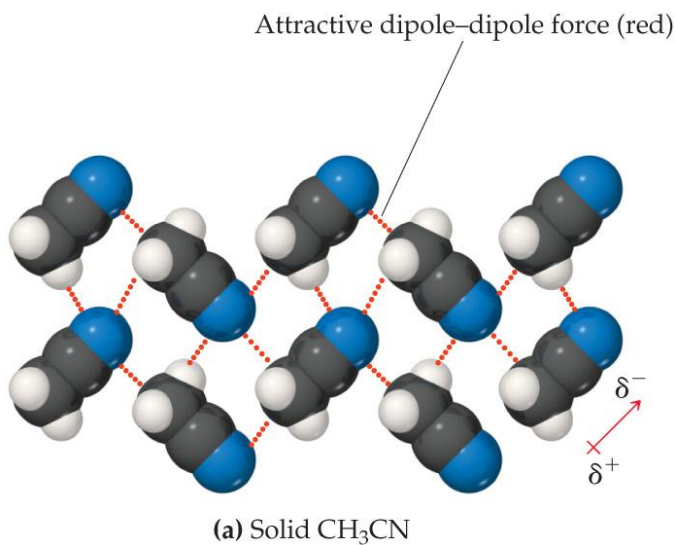
Polarizability & Boiling Point

- If something is **more difficult** to polarize, it has a *lower* boiling point.
- Remember: This means *less* intermolecular force (smaller molecule: lower molecular weight, fewer electrons).

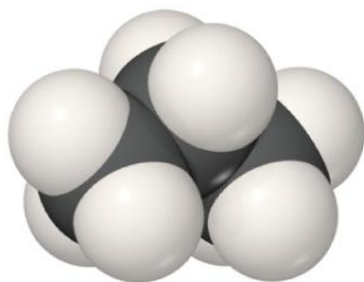


Dipole–Dipole Interactions

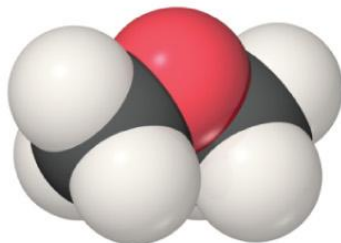
- Polar molecules have a more positive and a more negative end—a dipole (two poles, δ^+ and δ^-).
- The oppositely charged ends attract each other. 相反的



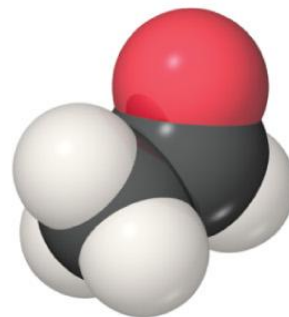
Dipole–Dipole Interactions



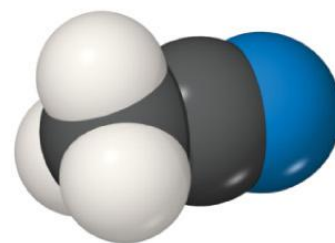
Propane
CH3CH2CH3
MW = 44 amu
 $\mu = 0.1 \text{ D}$
bp = 231 K



Dimethyl ether
CH3OCH3
MW = 46 amu
 $\mu = 1.3 \text{ D}$
bp = 248 K



Acetaldehyde
CH3CHO
MW = 44 amu
 $\mu = 2.7 \text{ D}$
bp = 294 K



Acetonitrile
CH3CN
MW = 41 amu
 $\mu = 3.9 \text{ D}$
bp = 355 K

Increasing polarity
Increasing strength of dipole–dipole forces

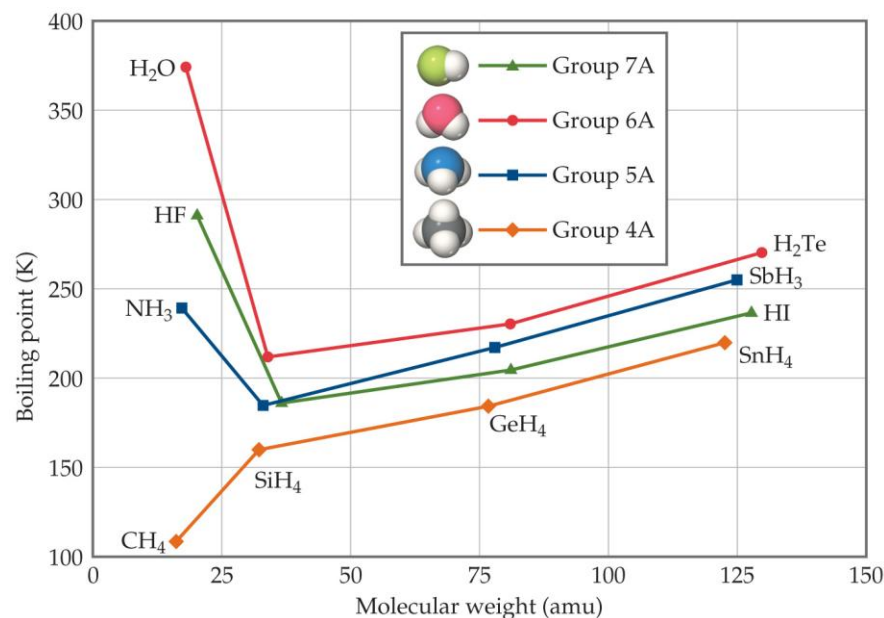
For molecules of approximately equal mass and size, the more polar the molecule, the higher its boiling point.

Which Have a Greater Effect: Dipole–Dipole Interactions or Dispersion Forces?

- If two molecules are of comparable size and shape, dipole–dipole interactions will likely be the dominating force.
- If one molecule is much larger than another, dispersion forces will likely determine its physical properties.

What Does This Graph Show Us?

- In a group, the period 3/4/5 elements have higher boiling points as the group member gets larger.
- What happens with the period 2 elements? For group 4A, the trend is continued. What about for the other groups?

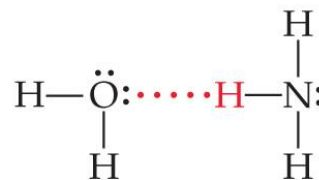
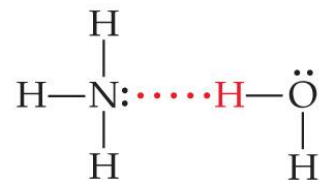
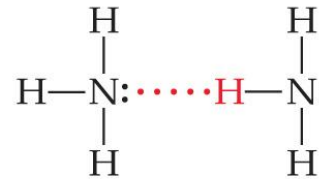


Hydrogen Bonding

- The dipole–dipole interactions experienced when H is bonded to N, O, or F are unusually strong.
- We call these interactions **hydrogen bonds**.
- A hydrogen bond is an attraction between a hydrogen atom attached to a highly electronegative atom and a nearby small electronegative atom in another molecule or chemical group.

Covalent bond,
intramolecular

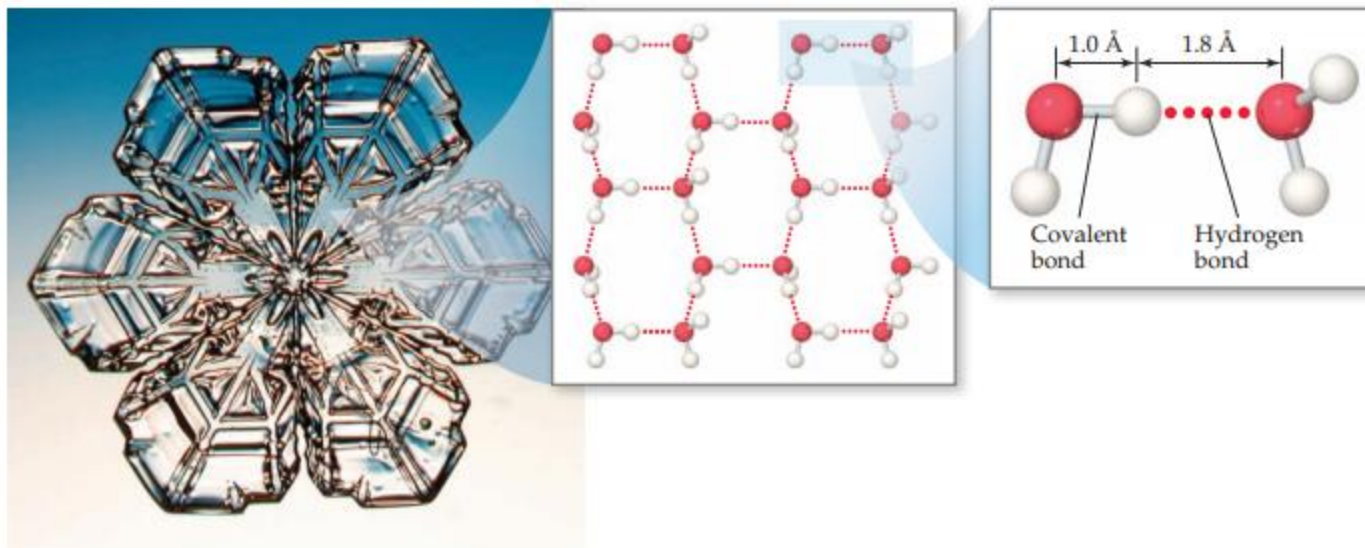
Hydrogen bond,
intermolecular



What Forms Hydrogen Bonds?

- Hydrogen bonding arises in part from the high electronegativity of nitrogen, oxygen, and fluorine.
- These atoms interact with a nearly bare nucleus (which contains one proton).

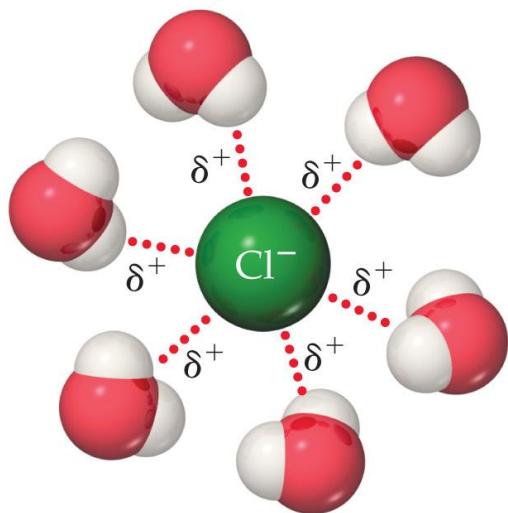




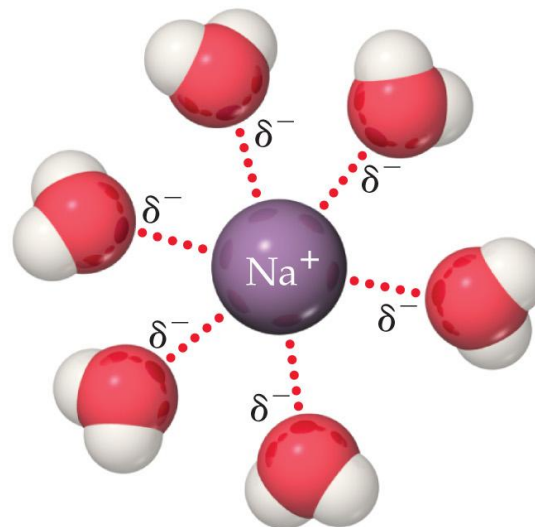
▲ **Figure 11.11** Hydrogen bonding in ice. The empty channels in the structure of ice make water less dense as a solid than as a liquid.

Ion–Dipole Interactions

- Ion–dipole interactions are found in solutions of ions.
- The strength of these forces is what makes it possible for ionic substances to dissolve in polar solvents.

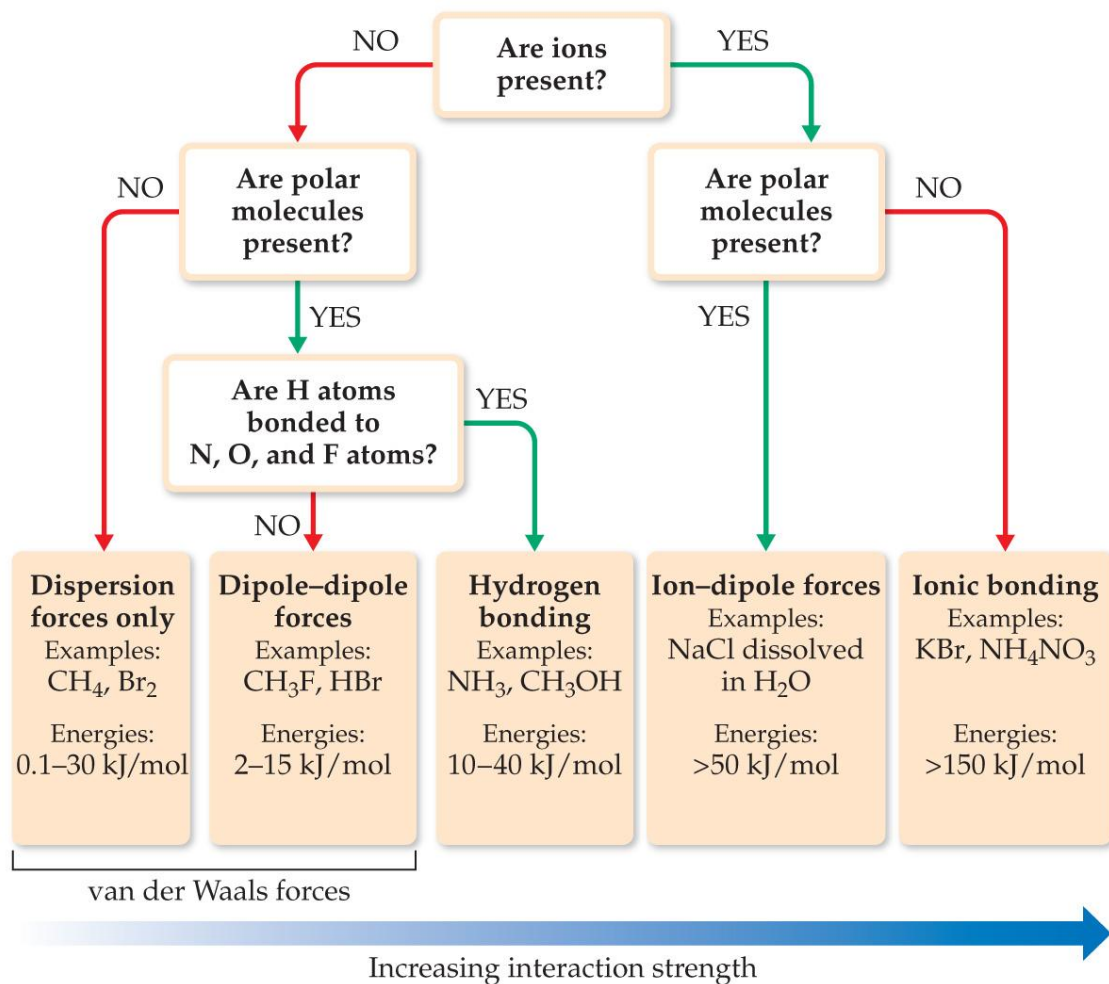


Positive ends of polar molecules are oriented toward negatively charged anion



Negative ends of polar molecules are oriented toward positively charged cation

Summarizing Intermolecular Forces



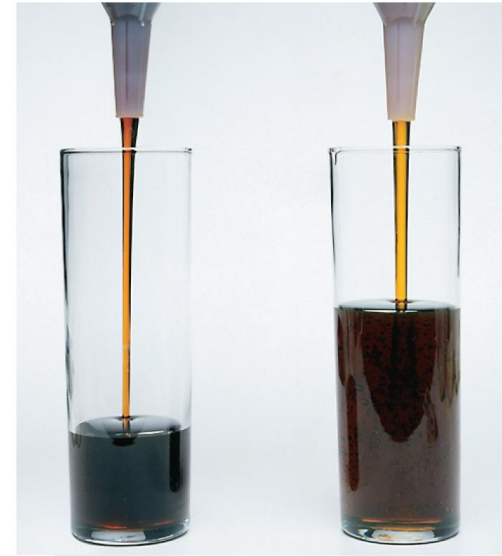
At which point in this flowchart would a distinction be made between SiH_4 and SiH_2Br_2 ?

Liquid Properties Affected by Intermolecular Forces

- boiling point and melting point
- viscosity
- surface tension
- capillary action

Viscosity

- Resistance of a liquid to flow is called **viscosity**.
- It is related to the ease with which molecules can move past each other.
- Viscosity increases with stronger intermolecular forces and decreases with higher temperature.



SAE 40
higher number
higher viscosity
slower pouring

SAE 10
lower number
lower viscosity
faster pouring

Table 11.5 Viscosities of a Series of Hydrocarbons at 20 °C

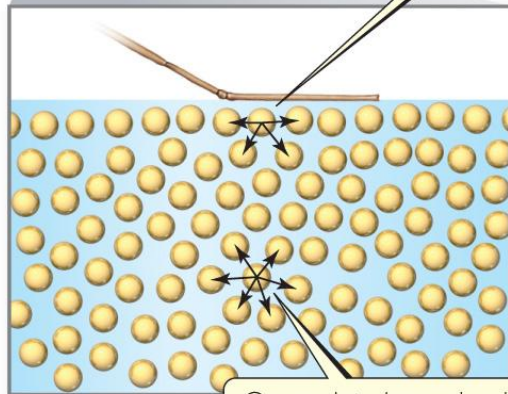
Substance	Formula	Viscosity (kg/m-s)
Hexane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	3.26×10^{-4}
Heptane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	4.09×10^{-4}
Octane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	5.42×10^{-4}
Nonane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	7.11×10^{-4}
Decane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	1.42×10^{-3}

Surface Tension

- Water acts as if it has a “skin” on it due to extra inward forces on its surface. Those forces are called the **surface tension**.



On any surface molecule, there is no upward force to cancel the downward force, which means each surface molecule “feels” a net downward pull



On any interior molecule, each force is balanced by a force pulling in the opposite direction, which means that interior molecules “feel” no net pull in any direction

<https://www.khanacademy.org/science/ap-biology/chemistry-of-life/structure-of-water-and-hydrogen-bonding/v/surface-tension>

Cohesion and Adhesion

- Intermolecular forces that bind similar molecules to one another are called *cohesive forces*.
- Intermolecular forces that bind a substance to a surface are called *adhesive forces*.
- These forces are important in capillary action.

Capillary Action

- The rise of liquids up **narrow** tubes is called **capillary action**.
- Adhesive forces attract the liquid to the wall of the tube.
- Cohesive forces attract the liquid to itself.
- Water has stronger adhesive forces with glass; mercury has stronger cohesive forces with itself.

